

product team and its supports have to trust Kickstarter. But with smart contracts we can build a similar system that doesn't require a third-party like Kickstarter. We can program the smart contract so that it holds all the received funds until a certain goal is reached. The supporters of a project can now transfer their money to the smart contract. If the project gets fully funded, the contract automatically passes the money to the creator of the project. If the project fails to meet the goal, the money automatically goes back to the supporters. As smart contracts are stored on a blockchain, everything is completely distributed. With this technique, no one is in control of the money.

Why should we trust a smart contract? because smart contracts are stored on a blockchain, they inherit some interesting properties. They are immutable and they are distributed. Being immutable means that once a smart contract is created, it can never be changed again. So no one can go behind your back and tamper with the code of your contract. And being distributed means that the output of your contract is validated by everyone on the network. So a single person cannot force the contract to release the funds because other people on the network will spot this attempt and mark it as invalid. Tampering with smart contracts becomes almost impossible. Smart contracts can be applied to many different things, not just on crowd-funding. Banks could use it to issue loans or to offer automatic payments. Insurance companies could use it to process certain claims. Postal companies could use it for payment on delivery, and so on. Right now there are a handful of blockchains who support smart contracts, but the biggest and well known one is Ethereum. It was specifically created and designed to support smart contracts. They can be programmed in a special programming language called Solidity. This language was specifically created for Ethereum and uses a syntax that resembles Javascript. It is worth noting that Bitcoin also has support for smart contracts although it's a lot more limited compared to Ethereum. So now we know what smart contracts are and what problem they solve.

When a product is discharged from a manufacturer, it has certain pre-conditions including temperature conditions, expected date of delivery, fragility etc. When we use blockchain for supply chain management, we make all these conditions a part of the smart contract which is to be strictly adhered to by all the intermediate carriers. This solves the problem of multiple peer to peer communication and makes blockchain a truly effective solution for supply chain management.